MIL-P-63345A (AR)
14 May 1984
SUPERSEDING
MIL-P-63345 (AR)
05 December 1980

MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804
METAL PARTS ASSEMBLY

This specification is approved for use the U.S. Army Armament Muntions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Metal Parts Assembly for the Projectile, Practice, 155MM, M804.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defence Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pre-treatment of Ferrous Surfaces for Original Coatings

MILIT*RY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Inspection by Attributes
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.12)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9331795 - Projectile, 155MM Practice, M804 Metal Parts Assembly
9331807 - Packaging and Marking for Pallet, Wood, Ammunition for Projectile 155MM Practice, M804
9354433 - Alternate Packing and Marking for Pallet, Wood, Ammunition, for Projectile, 155MM, Practice, M804

PUBLICATIONS

FRANKFORD ARSENAL

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the cxurrent DoDISS and the supplement thereto, if applicable.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, N.Y., N.Y. 10018).

ASTM E-8 - Tension Testing of Metallic Materials
ASTM E-10 - Brinell Hardness of Metallic Materials
ASTM B-117 - Salt Spray (Fog) Testing
ASTM A-711 - Carbon and Alloy Steel Blooms, Billets and Slabs for Forging
ASTM E-103 - Rapid Indentation Hardness Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 916 Race Street, Philadelphia, Pa., 19103).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Projectile assembly. The Projectile Assembly and components shall comply with all requirements specified on Drawing (Dwg.) 9331795 and with all requirements specified in applicable specifications.

3.3 Steel quality. The item contractor shall establish procedures to provide a forge quality steel in accordance with ASTM A711; i.e., the steel shall be free of pipe, porosity and heavy segregation. The manufacturer will provide a certification to cite the specific revisions of the industry standard under which the steel was manufactured.

3.4 Proving ground. The projectiles, when tested as specified, shall show no loss or break-up of metal components. The fuzed projectile when assembled with a smoke cannister shall on impact, produce a smoke signature visible across 4000 meters of level terrain.

3.5 Mechanical properties.

3.5.1 Body. Mechanical properties of the machined body as required by applicable drawing shall apply.

3.5.2 Rotating band. After application and subsequent removal the rotating band shall have the following mechanical properties:

a. Tensile Strength - 206.8 Megapascals (MPa) (30,000 psi) minimum (min)
b. Elongation - 25 percent in 20 millimeters (mm) or .75 inch (in) of gage length min.

3.6 Rotating band.

3.6.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

3.6.2 Band blank - (Gilding metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification; except that when hot banding at not less than 427 degrees Celsius (^{o}C), no prior annealing is necessary.

3.6.3 Application. The bands may be applied by either hot or cold process but if applied hot, the bands shall not be heated above 870 degrees Celsius (^{o}C) to prepare them for application to the body. The bands shall be heated uniformly by a process which will prevent oxidation (loose scale) and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.6.4 Band seating sidewall contact. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such a manner so as not to distort the body wall. In addition, there shall be no surface gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimension.

3.6.5 Band seating tightness. The diametrical clearance between the rotating band and the band seat shall not exceed 0.15mm (.006 in), when the destructive method of testing is used or 0.23mm (.009 in) when non-destructive method of testing is used.

3.7 Metal defects.

3.7.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects.

3.7.2 Visual standards. Visual standards for evaluation of metal defects will be established by the contractor with guidance from the Government Agency for each contract after the first months production. However, establishment of visual standards will not waive the requirements for compliance with 3.7.1.

3.8 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Tool control dimensions. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

3.10 Workmanship. The requirements for workmanship are as shown on the applicable drawings and specifications and the following:

3.10.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and specifications. Cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.10.2 Painting. Painting shall comply with the requirements of the applicable drawings and specifications. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.10.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determinations.

3.10.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.10.5 Burr. No part shall have a burr which might interfere with the assembly or function of the projectile or which might be injurious to personnel handling the item.

3.10.6 Foreign matter. No part or assembly shall contain dirt, grease, rust, corrosion or other foreign matter.

3.10.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First artcicle inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Body Assembly	9331299	5
Blank Rotating Band	7548993	5
Body	9331797	5
Metal Parts Assembly	9331795	20

All assemblies and components shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table I. All inspections listed in Table I shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Projectile Assemblies and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM, Practice M804, Metal Parts Assembly (Dwg. 9331795)				
	Examination for Defects (Except Proving Ground Test)	15		3.2	4.4.2
	Proving Ground Test	20		3.4	4.5.5

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Heat treatment inspection lots of projectiles shall be defined as follows:

4.4.1.1 Standard procedure. A heat treatment lot of bodies shall consist of bodies from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process.

4.4.1.2 Miscellaneous lot. Bodies manufactured from small heats of steel, or bodies that have been separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000 bodies.

4.4.1.3 Use. The heat treatment lot constituted per 4.4.1.1 or 4.4.1.2 shall be the basis for selecting samples for mechanical properties tests. Bodies in each heat treatment lot shall be identified to that particular lot until satisfactory mechanical properties tests have been completed.

4.4.1.4 Projectile lot. Heat treatment lots that have been determined to have complied with yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.1	Body		9331797
			NEXT HIGHER ASSEMBLY 9331299

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Metal defective		100%	3.7	Visual
102	Runout of rear face		0.40%	3.2	Gage
103	Depth of scoring		0.40%	3.2	Visual
104	Scoring or knurling missing from rotating band seat		0.40%	3.2	Visual
105	Runout of pitch diameter of thread		0.40%	3.2	Gage
106	Runout of ogive		0.40%	3.2	Gage
107	Runout of bourrelet		0.40%	3.2	Gage
108	Runout of boattail		0.40%	3.2	Gage
109	Variation in wall thickness		0.40%	3.2	Gage
110	Pitch diameter of thread, max.		0.40%	3.2	Gage
111	Minor diameter of thread, max.		0.40%	3.2	Gage
112	Length of effective thread, min.		0.40%	3.2	Gage
113	Mechanical properties of body		4.4.3.1	3.5.1	4.5.2
114	Length of effective thread, max.		0.40%	3.2	Gage
Minor					
201	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
202	Surface finish improper		0.65%	3.10.3	Visual
203	Evidence of shot or sandblasting on rotating band seat		0.65%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9331797
					NEXT HIGHER ASSEMBLY 9331299
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
204	Burr		0.65%	3.10.5	Visual
205	Diameter of thread chamfer		0.65%	3.2	Gage
206	Diameter at rear of boattail		0.65%	3.2	Gage
207	Length from front face to rear of bourrelet		0.65%	3.2	Gage
208	True position of holes		0.65%	3.2	Gage
209	Diameter of holes, max.		0.65%	3.2	Gage
210	Diameter of rotating band seat		0.65%	3.2	Gage
211	Location of rotating band seat		0.65%	3.2	Gage
212	Width of rotating band seat		0.65%	3.2	Gage
213	Length, total		0.65%	3.2	Gage
214	Length from front of boattail to rear of rotating band seat		0.65%	3.2	Gage
215	Thickness through base		0.65%	3.2	Gage
216	Diameter of bourrelet, min.		0.65%	3.2	Gage
217	Diameter of body		0.65%	3.2	Gage
218	Length from front face to forward basic diameter		0.65%	3.2	Gage
219	Length from front face to rear basic diameter		0.65%	3.2	Gage
220	Thickness of body wall, min.		0.65%	3.2	Gage
221	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.2	Body Assembly				9331299
					NEXT HIGHER ASSEMBLY
					9331795
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Runout of rotating band		0.40%	3.2	Gage
102	Diameter of rotating band		0.40%	3.2	Gage
103	Location of rotating band		0.40%	3.2	Gage
104	Excessive gap between rotating band and sidewall		4.4.3.2	3.6.4	4.5.3.2
105	Excessive gap between rotating band and seat		4.4.3.3	3.6.5	4.5.3.1
106	Rotating band mechanical properties		4.4.3.2	3.5.2	4.5.3.2
107	Rotating band seat not clean, prior to application		0.40%	3.6.1	Visual
Minor					
201	Protective coatings or paint missing, incomplete, or improper		0.65%	3.2	Visual
202	Marking missing		0.65%	3.2	Visual
203	Length, max.		0.65%	3.2	Gage
204	Diameter of rotating band groove		0.65%	3.2	Gage
205	Length from rotating band groove to start of angle		0.65%	3.2	Gage
206	Length of rotating band flat		0.65%	3.2	Gage
207	Angle at front of rotating band		0.65%	3.2	Gage
208	Angle at rear of rotating band relief groove		0.65%	3.2	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER 9331299
4.4.2.2	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9331795
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
209	Width of rotating band		0.65%	3.2	Gage
210	Width of rotating band groove		0.65%	3.2	Gage
211	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
212	Width of rotating band relief groove		0.65%	3.2	Gage
213	Evidence of poor workmanship		0.65%	3.10	Visual
214	Assembly weight incorrect		0.65%	3.2	Scale

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.3	Projectile, 155MM, Practice, M804 Metal Parts Assembly				9331795
					NEXT HIGHER ASSEMBLY 9331794
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rotating band damaged, prior to assembly of grommet		0.40%	3.10.7	Visual
102	Salt spray test		4.4.3.4	3.10.2	4.5.4
103	Protective coating thickness		4.4.3.4	3.10.2	4.5.4
104	Diameter of bourrelet, max.		0.40%	3.2	Gage
Minor					
201	Corrosion preventive compound missing or improperly applied		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER 9331807
4.4.2.4	Packaging and Marking				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet cover damaged or improperly assembled		0.40%	3.2	Visual
102	Strapping or wire missing or loose		0.40%	3.2	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping or wire improperly engaged		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Mechanical properties of body.

4.4.3.1.1 Hardness. In order to determine uniformity of heat treatment, each body in the heat treatment lot shall be subjected to a hardness test approved by the procuring activity.

4.4.3.1.2 Tension test. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.4.3.2 Rotating band mechanical properties and band seating sidewall contact. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation to determine compliance with applicable requirements. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Band seating tightness.

4.4.3.3.1 Non-destructive method. Each body assembly shall be subjected to this test. Failure to comply with applicable requirements shall be cause for rejection of that body assembly.

4.4.3.3.2 Destructive method. When the equipment for the nondestructive method of testing is not available, one body assembly shall be selected from each 8-hour production or fraction thereof per banding machine in operation. Failure of any band to comply with applicable requirements shall be cause for rejection of the production represented by the sample.

4.4.3.4 Protective coating - salt spray. The sample size and acceptance criteria shall be as specified in the applicable specification. If failure occurs, all items processed since last acceptance shall be rejected and corrective action taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

4.4.3.4.1 Protective coating - thickness. Two samples shall be selected at random to represent each four hour period, or less, of continuous paint production. The sample projectiles shall be inspected for paint coating thickness at a minimum of four separate locations on each projectile. Any projectile failing to meet the applicable requirement shall be classed defective and every projectile painted since the last acceptable sample shall be rejected.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Methods of inspection.

4.5.1 Metal defects. The projectile shall be inspected visually using peripheral lights to assure conformance with the applicable requirements. Visual standards, in accordance with applicable requirements shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.5.2 Mechanical properties of body.

4.5.2.1 Hardness. Prior to final machining, a Brinell hardness test shall be performed on the outside surfaces at the band seat location or at any other area on the body deemed necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches mimimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification shall be performed using Brinnel test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 105MM Brinnel ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000Kg. The amount of penetration of the indentor (105MM ball) during that time period when the major load is applied including the dwell time at

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.2.2 Tension test. From the sample projectiles selected, two tension specimens conforming to the largest obtainable standard round type specimens shall be tested in accordance with the procedure prescribed by ASTM E-8. The specimens shall be taken longitudinally, from opposite sides of the body in such a manner that the center of the gage length of the machine test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their locations are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM E-8.

4.5.2.2.1 Base section. The base section tensile specimen shall be taken transversely, 20mm (.75 in) from the center line. Failure to meet the yield strength or elongation requirements shall require a retest from the remaining portion of the base section. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the inspection lot.

4.5.2.2.2 Nose section. One longitudinal specimen shall be taken from the ogive, not more than 76mm (3 in) from front face. Failure to meet the yield strength or elongation requirement shall be cause for rejection of the inspection lot.

4.5.3 Rotating band.

4.5.3.1 Band seating tightness.

4.5.3.1.1 Nondestructive method. Each sample shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 2.1), for test of 155MM, Proj., M107.

4.5.3.1.2 Destructive method. The body assembly selected should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured at 3 locations, 60 degrees apart, on the points along the cylindrical

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurements shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.3.2 Mechanical properties and sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact.

4.5.3.2.1 Mechanical properties. The section provided shall be flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having two centimeter gage length to accomodate the tensile machine, shall be removed from the blank band. The specimen shall be tested in accordance with the procedure prescribed by ASTM Method E-8.

4.5.3.2.2 Sidewall contact. The specimen provided shall be sectioned into two parts exposing four surfaces, each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements.

4.5.4 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon.

4.5.4.1 Salt spray testing. Salt spray testing shall be accomplished on production bodies or sectioned bodies that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned bodies are used, sectioning shall be accomplished to allow testing of all major areas of the body on an alternating basis. The Salt spray test shall be performed utilizing the equipment and procedures specified by ASTM-B117 and TT-C-490.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

4.5.4.2 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially available equipment.

4.5.5 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. Each metal parts assembly shall have a liner and smoke canister assembled. The sample projectiles selected shall be fired in a M185 or M199 series cannon using a M119A1, Zone 8 charge. Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 20° Celsius (70°F) plus or minus 1° Celsius (2°F) for a minimum of 24 hours prior to testing. Pressure-time traces and differential pressures shall be obtained for information only by means of a spindle tap in the breech of the cannon and a piezo gage at the projectile base location for each projectile fired. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes. An observer shall be stationed 4000 ± 200 meters from the edge of the impact area to record presence of smoke signature.

5. PACKAGING

5.1 Packing and marking. The projectile shall be packed in accordance with Dwg. 9331807 or 9354433 and MIL-STD-1169.

6. NOTES

6.1 Intended use. The projectile covered by this specification is intended for use as a target practice round for 155MM Howitzers.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAR-I (D), Dover, New Jersey 07801.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.6 Submission of ballistic test data and ammuntion data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammuntion data cards shall be forwarded to Commander, ARDC, ATTN: DRSMC-QAR-Q (D), and DRSMC-LCU-SA (D), Dover, New Jersey 07801 and DRSMC-QAD-D(R) Army Master Depository, Rock Island IL 61299.

6.7 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.8 Band seating. The operating procedures for use of Detroit Testing Machine as prepared by the Inspection Equipment Branch, U.S. Army, Frankford Arsenal, DRDAR-FA.

6.9 Steel grade. It is the responsibility of the manufacturer of the projectile metal parts to select a fully-kilned steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions and mechanical properties as an end item requirement.

6.10 Raw material. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade selected by the projectile manufacturer and complying with specification ASTM designation A-711.

6.11 Variables. In the application of steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.

weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices and processing techniques having capabilities of compensating for such variables.

6.12 Drawings. Drawings listed in Section 2 of the specification under the heading US Army Armament Research and Development Center (ARDC), may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A810)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:42Z
Check the source to verify that this is the current version before use.